# Ramim Ali Siddiqui

*AI/ML Engineer*

+92 322-9313588 · ramimalisiddiqui0@gmail.com · Karachi, Pakistan
ramim-siddiqui.vercel.app · github.com/Ramim-Siddiqui · linkedin.com/in/ramim-siddiqui

## Professional Summary

AI/ML engineer who builds LLM agents, document AI and computer vision services in Python and ships them as tested, deployed APIs. At Gems Pathways I built the MCQ, photo-validation and data-entry tools used with the company's learning portal; on my own, a voice agent that registers patients by phone, with 101 automated tests, running live.

## Skills

| | |
|---|---|
| **LLMs & agents** | LangGraph, LangChain, retrieval-augmented generation, tool calling, voice agents (Vapi), Gemini, OpenAI and Groq APIs |
| **ML & vision** | PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, MediaPipe, ONNX Runtime |
| **Backend & data** | FastAPI, Flask, Playwright, PostgreSQL, MongoDB, Qdrant, Boomi, Postman (OData APIs) |
| **Deployment** | Docker, Linux servers, GitHub Actions, Vercel, Hugging Face Spaces, MLflow, AWS and GCP (university) |
| **Languages** | Python, SQL, JavaScript |

## Professional Experience

**AI Engineer**, *Gems Pathways, Karachi* — Jul 2026 – Present

- **MCQ Studio** Code: built the FastAPI service behind the company's learning portal that writes chapter-weighted MCQs from textbooks and extracts MCQs from question papers; Qdrant retrieval, five LLM providers, 173 automated tests, Docker.
- **PortraitPass** Live · Code: built the service that validates student ID photos with MediaPipe (one person, frontal head, no full-body shot, no blur) and cuts out the background with BiRefNet.
- **Smart Form Filler** Code: built a Playwright tool that replays one recorded data-entry session for every row of an Excel sheet, with per-row checks, resumable runs and parallel browsers.
- Integrate these services with the company's Laravel portals together with the backend and frontend teams.

**Junior Solution Engineer**, *Squarera, Karachi* — Oct 2025 – Jun 2026

- Built and automated enterprise ETL integrations between business systems in Boomi.
- Tested and debugged OData APIs with Postman while building the integrations.
- Optimised SQL transformations for faster queries and analytics-ready datasets.

**Freelance Data Scientist**, *Online Dimensions, remote* — Jul 2024 – Jun 2025

- Built NLP and recommendation models for client projects and tracked their performance with MLflow.
- Automated data-preprocessing pipelines over 500k+ records.
- Served predictions through Flask and FastAPI endpoints with responses under 200 ms.

**Web Development Intern**, *Future Power Company, Karachi* — Jul 2024 – Aug 2024

- Built the company website with React and Node.js, with contact-form email through Nodemailer.

## Projects

**Voice AI Patient Registration Agent** — Live · Code

Phone agent that registers patients in natural conversation: five backend tools, validation shared with the REST API, a live call dashboard and 101 automated tests. *FastAPI, Vapi, PostgreSQL, Vercel*

**Support Ticket Resolution Agent** — Live · Code

LangGraph workflow that classifies tickets, retrieves knowledge with Gemini embeddings, drafts replies and reviews them against rules, escalating after two failed drafts; about 2 s warm response. *LangGraph, Gemini, FastAPI*

**Smart Attendance System** — Code

Face-recognition attendance from CCTV on a CPU-only PC: SCRFD and ArcFace on ONNX Runtime, one IN/OUT log per visit, numbered unknown visitors, automatic reconnection. *OpenCV, ONNX Runtime, Tkinter*

**Source-Code Vulnerability Detection with Graph Neural Networks** — Code

Final-year research: GraphSAGE over the syntax trees of 27,318 Devign C/C++ functions; 93.0% accuracy and macro F1 0.93 on a positional hold-out with near-duplicates kept, so an in-distribution estimate, not a benchmark result. *PyTorch Geometric, libclang*

## Education

**B.S. Computer Science**, Sukkur IBA University, CGPA 3.16 / 4.00 — 2021 – 2025

**F.Sc. Pre-Engineering**, Govt. Islamia Science College — 2019 – 2021

## Achievements

Winner, Robo War Competition, SIBA Test, Sukkur IBA University (2025); second place, Cyber Security Hackathon Workshop organised by Ignite.